MEMO — Milestone 3: Skew-t Refinement & ML Challenger Evaluation

Swiss Market Forecasting Project | UBSG GJR-GARCH skew-t vs. symmetric-t; GJR-GARCH vs. LightGBM challenger

Three findings below carry equal weight — a positive result, a negative-but-informative result, and the process record — none is a headline over the others.

KEY FINDINGS

Item	Finding	Evidence
Skew-t adopted for UBSG	Hansen's skew-t replaces symmetric-t as UBSG's production innovation distribution. In-sample LR test favors skew-t ($p = 0.0034$), AIC agrees, BIC is a near-wash (disclosed, not smoothed over). Out-of-sample walk-forward coverage improves: 99% VaR Kupiec p moves $0.014 \rightarrow 0.079$, out of the borderline zone. Verified via a paired day-by-day comparison: 3 days reclassified favorably (borderline-under-symmetric-t → comfortably-clean-under-skew-t), 0 reclassified unfavorably, no clustering (span Mar 2023 – Sep 2025). Mechanism is mostly, not purely, skewness: 2 of the 3 flipped refits show ν barely moving (β=driven by λ), but refit 10 shows ν dropping 2.1 points alongside λ — a joint tails-fatten-and-skew effect, disclosed explicitly rather than folded into a clean "skewness fixed it" line.	LR test; walk-forward
ML challenger: genuine strengths, does not replace champion	LightGBM (variance regression + direct quantile regression, same walk-forward window) evaluated against a criterion fixed before fitting: beat champion on QLIKE AND do not fail calibration where champion passes. Result: QLIKE -7.078 (challenger) vs -7.017 (champion) — challenger wins. 95% Kupiec: challenger $p = 0.532$ (clean) vs champion $p = 0.092$ — challenger wins. 99% Kupiec: challenger $p = 0.026$ (fails) vs champion $p = 0.079$ (passes) — challenger fails the pre-registered condition, exactly where it matters most for tail risk. Per the rule fixed in advance: champion retains status. Caveat disclosed without moving the decision: under the same multiple-testing lens used throughout this project, the challenger's 99% failure would not survive correction either — tempers confidence in the loss, does not reverse it. Challenger retained as a documented, monitored alternative, not discarded.	Walk-forward backtest
Process record (dead ends included)	Session reset wiped the entire local project mid-conversation; rebuild repeated the exact mkdir brace-expansion failure from Milestone 1 kickoff (fix hadn't been saved as a script — now is, `setup_scaffold.sh`). Rebuilt pipeline cross-checked against Milestone 2's already-validated numbers before trusting any new result (symmetric-t 99% Kupiec reproduced exactly: 24 violations, $p = 0.0144$). Paired comparison and ν/λ decomposition run specifically to avoid re-asserting an unverified mechanism claim, the same failure mode as an earlier milestone's "differentiated repricing" and "refit-8 crisis" narratives that were built and then killed by their own evidence.	Cross-check vs M2

LIMITATIONS

Skew-t mechanism attribution rests on a 3-observation paired sample — directionally clear, not a large-sample claim. ML challenger uses one feature set and default-adjacent hyperparameters, untuned; QLIKE gap could move with further development. Both findings inherit Milestone 2's standing caveats (short post-merger UBSG sample, deferred event-clustering methodology) — tracked separately in the standing open-threads note, not repeated here.

NEXT STEP

Skew-t GJR-GARCH is UBSG's production specification. ML challenger retained as monitored alternative — revisit if QLIKE gap persists/widens with more data, or 99% calibration improves with feature/hyperparameter development. Milestone 3 closes. Standing open-threads note follows separately, per instruction.

Full detail and code: `notebooks/03_skewt_ml_challenger.ipynb`.